Tawhid Bhuiyan

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Research Interests

Ph.D. student at Columbia University working on **compilers** and **hardware/software co-design**: profile-guided optimization for data center applications and energy efficiency of GPUs and machine learning systems. Seeking research internships for Summer 2027.

Education

Columbia University — *New York, USA*
Ph.D. in Electrical Engineering (Advisor: Prof. Tanvir Ahmed Khan) — Sept. 2024 – May 2030 (expected)

Bangladesh University of Engineering and Technology (BUET) — *Dhaka, Bangladesh*
M.Sc. in Computer Science and Engineering — July 2021 – Aug. 2024

Bangladesh University of Engineering and Technology (BUET) — *Dhaka, Bangladesh*
B.Sc. in Computer Science and Engineering ▪ *Ranked 1st of 142 students* — Mar. 2016 – Feb. 2021

Research Experience

Columbia University — *New York, USA*
Graduate Research Assistant — Sept. 2024 – Present

- **Optimizing data center applications with stale profiles (Wax, ASPLOS '26):** Designed a technique that uses debug and source-code information to apply stale profiles to fresh binaries. Achieved 5.76–26.46% speedups on 5 data center applications (gcc, clang, MySQL, PostgreSQL, MongoDB), 1.20–7.86% more than the state of the art, recovering 65–93% of fresh-profile benefits.
- Implemented on top of LLVM BOLT and extended the approach to Propeller, raising clang's speedup from 5.4% to 6.4%. Presented the work in invited talks at Google and Meta.
- **Reverse engineering hardware-managed DVFS (HOST '25):** Built a Linux kernel module for low-overhead, high-frequency sampling of model-specific registers (MSRs) to reconstruct an Intel processor's DVFS mechanism, uncovering undocumented MSR bits. Demonstrated a model-stealing attack with ML proxy models (99% R^2, up to 92% F1).
- **Energy efficiency of GPUs and ML systems (ongoing):** Characterizing and modeling the energy of LLM workloads on data center GPUs using architectural simulators and hardware profiling tools.

Publications

[ASPLOS '26] Tawhid Bhuiyan, Sumya Hoque, Angélica Moreira, Tanvir Ahmed Khan.
Wax: Optimizing Data Center Applications With Stale Profile. *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Pittsburgh, PA, 2026.* [Code]

[HOST '25] Ryan Piersma, **Tawhid Bhuiyan**, Tanvir Ahmed Khan, Simha Sethumadhavan.
Reverse Engineering DVFS Mechanisms. *IEEE International Symposium on Hardware Oriented Security and Trust (HOST), San Jose, CA, 2025.*

[TCS '22] Mohammad Tawhidul Hasan Bhuiyan, Muhammad Rashed Alam, M. Sohel Rahman.
Computing the Largest Common Almost-Increasing Subsequence. *Theoretical Computer Science, Vol. 930, 2022.*

[IJERPH '21] Mohammad Tawhidul Hasan Bhuiyan, Irtesam Mahmud Khan, Sheikh Saifur Rahman Jony, Renee Robinson, Uyen-Sa D. T. Nguyen, David Keellings, M. Sohel Rahman, Ubydul Haque.
The Disproportionate Impact of COVID-19 among Undocumented Immigrants and Racial Minorities in the US. *International Journal of Environmental Research and Public Health, 18(23), 2021.*

Talks

Jun. 2026 *Enabling Energy Efficiency for Distributed Machine Learning*, ASTRA-sim Tutorial at ISCA 2026
Mar. 2026 Wax: *Optimizing Data Center Applications With Stale Profile*, ASPLOS 2026, Pittsburgh, PA
Mar. 2026 Wax: *Optimizing Data Center Applications With Stale Profile*, invited talk at **Google**
Feb. 2026 Wax: *Optimizing Data Center Applications With Stale Profile*, invited talk at **Meta**
May 2025 *Reverse Engineering DVFS Mechanisms*, HOST 2025, San Jose, CA

Technical Skills

Compilers & PGO: LLVM, BOLT, Propeller, AutoFDO
GPU & Performance: CUDA, perf, NVIDIA Nsight, NVML, CUPTI, PyTorch Profiler
Architectural Simulators: Accel-sim, ASTRA-sim
Languages: C/C++, Python, Java, SQL, Bash, JavaScript
Libraries: PyTorch, Polars, Numba, Pandas, NumPy, Scikit-Learn, TensorFlow, Matplotlib

Teaching & Mentoring

Columbia University ▪ *Mentor* *Sept. 2024 – Present*

- *EECS 6894, Fall 2026:* Mentoring 15 students across 5 research projects.
- *EECS 6894, Fall 2025:* Mentored 15 students across 4 research projects, guiding teams through the design, implementation, and debugging of complex systems.
- Mentored Wenxuan Xu (M.S. student) in research on optimizing vLLM for energy efficiency, and Amir Zarandi (undergraduate) in research on GPU energy modeling.

BUET ▪ *Lecturer, Department of CSE* *Sept. 2021 – Aug. 2024*

- Developed curricula and taught Operating Systems (including hands-on kernel modification), Database Systems, Software Development, Computer Graphics, Computer Programming, and Digital Logic Design.

Brac University; United International University ▪ *Lecturer* *Feb. 2021 – Sept. 2021*

Awards & Service

- BUET Alumni Association Award for the top position in the department (2022)
- Dean's Honor List, BUET, all 8 undergraduate semesters; University Merit Scholarship, BUET (2016–2021)
- Organizing Committee, International Conference on Networking, Systems and Security (NSysS) 2021, 2022, 2023